

CHAPTER 3

Interaction of Light with Matter

Main content

➤ 3.1 Macroscopic Aspects for Solids

First we consider matter as a homogeneous medium described by the complex dielectric function $\epsilon(\omega)$ or by the complex index of refraction $\tilde{n}(\omega)$

➤ 3.2 Microscopic Aspects

We will discuss the interaction of the radiation field with individual atoms. In this case quantum mechanics must be used.

3.1 Macroscopic Aspects for Solids

3.1.1 Boundary Conditions

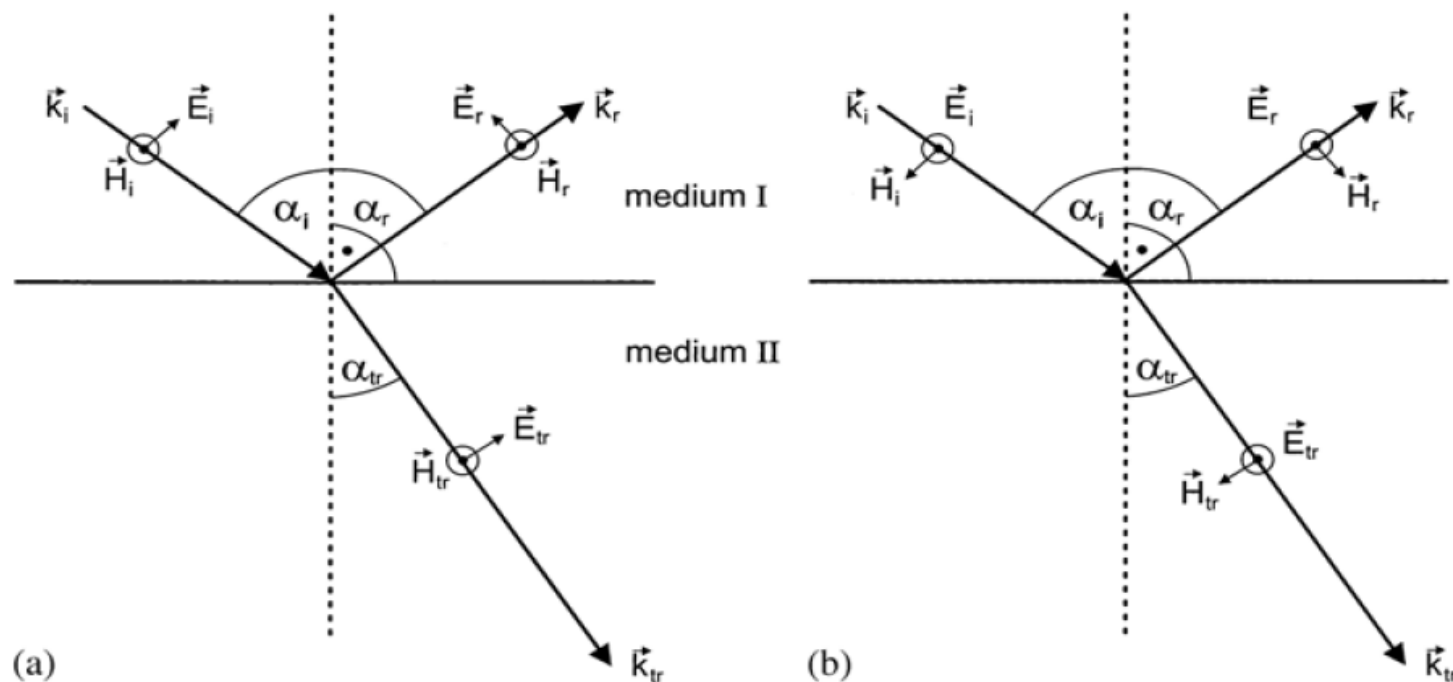


Fig. 3.1. The $\mathbf{E}$ and $\mathbf{H}$ fields and the wave vectors $\mathbf{k}$ for incident, transmitted and reflected beams at an interface between two isotropic media for two different, orthogonal, linear polarizations (a) and (b) respectively

- Assume :1. media I and II are isotropic ; 2. there is only one reflected and one transmitted beam
- The coefficients of reflection r and transmission t of the interface :
- For Fig.3.1a :

$$r_{\parallel} = \frac{E_{0r}}{E_{0i}} \quad t_{\parallel} = \frac{E_{0tr}}{E_{0i}}$$

- For Fig.3.1b:

$$r_{\perp} = \frac{E_{0r}}{E_{0i}} \quad t_{\perp} = \frac{E_{0tr}}{E_{0i}}$$

- For measurement , reflectivity R and transmittivity T of an interface for the intensities:

$$R_{\perp, \parallel} = |r_{\perp, \parallel}|^2$$



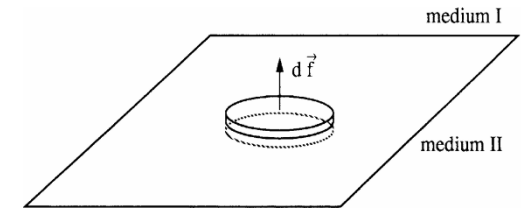
How to calculate them?

1. Maxwell's equations with boundary conditions
2. The laws of Gauss and Stokes

$$\int_{\text{volume}} \nabla \cdot \mathbf{A} \, d\tau = \oint_{\text{surface}} \mathbf{A} \cdot d\mathbf{f} \quad \int_{\text{surface}} (\nabla \times \mathbf{A}) \cdot d\mathbf{f} = \oint_{\text{line}} \mathbf{A} \cdot d\mathbf{s}$$

$$\int_{\text{volume}} \nabla \cdot \mathbf{A} d\tau = \oint_{\text{surface}} \mathbf{A} \cdot d\mathbf{f}$$

Fig. 3.2 Schematic drawing of the infinitesimally small cylinder used to deduce with (3.4) the boundary condition (3.6)



- Choose the integration volume as show Fig.3.2

$$\int_v \nabla \cdot \mathbf{D} d\tau = \oint_{\text{surface}} \mathbf{D} \cdot d\mathbf{f} = \int_v \rho d\tau$$

- Since we assumed no free charges: $\rho = 0$ $(\mathbf{D}_I - \mathbf{D}_{II}) \cdot d\mathbf{f} = (D_{n,I} - D_{n,II})df = \rho_s df = 0$

We obtain

$$D_n^I = D_n^{II} \quad (3.6a)$$

- The same way :

$$\int_{\text{surface}} (\nabla \times \mathbf{A}) \cdot d\mathbf{f} = \oint_{\text{line}} \mathbf{A} \cdot d\mathbf{s}$$

$$B_n^I = B_n^{II} \quad (3.6b)$$

$$E_t^I = E_t^{II} \quad (3.7a)$$

$$H_t^I = H_t^{II} \quad (3.7b)$$

- (3.6) and (3.7) represent the boundary conditions for electric and magnetic fields
- Must be applied to a specific problem:

Using the Maxwell's equation with these set of equations allows one to calculate the properties of the reflected and transmitted beams.

• 3.1.2 Laws of Reflection and Refraction

• All three beams have the same frequency: $\omega_i = \omega_r = \omega_{tr}$ (3.9)

• The laws of reflection and Snell's law of refraction:

$$\alpha_i = \alpha_r \quad (3.10a)$$

$$\mathbf{k}_i, \mathbf{k}_r, \mathbf{e}_n \text{ are in one plane.} \quad (3.10b)$$

• Also reads :

$$\frac{\sin \alpha_i}{\sin \alpha_{tr}} = \frac{n_{II}}{n_I} \quad \mathbf{k}_i, \mathbf{k}_r, \mathbf{e}_n \text{ are in one plane.} \quad (3.10c)$$

• When $n_I > n_{II}$, there is a critical angle α_i^c for which $\alpha_{tr} = 90^\circ$

$$\alpha_i^c = \arcsin \frac{n_{II}}{n_I} \quad (3.11)$$

For $\alpha_i \geq \alpha_i^c$, a totally reflected beam but no transmitted one: total reflection

Fig.3.3a : The distance of a few wavelengths

Fig.3.3b: When thickness of medium II $\sim \text{nm}$ and covered by material I : Attenuated, or frustrated, total reflection or the optical tunnel effect

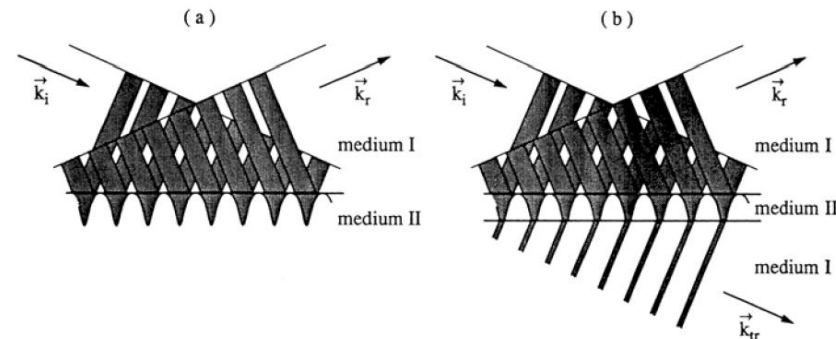


Fig. 3.3. The evanescent wave in the case of total internal reflection (a) and the arrangement for the “optical tunneling” effect or attenuated total reflection (b)

• 3.1.4 Reflection and Transmission at an Interface and Fresnel's Formulae

- **Weak absorption:** $|\kappa| \ll |n|$
- The Fresnel formulae are, for the regime of weak absorption.

$$r_{\perp} = \frac{n_{\text{I}} \cos \alpha_i - n_{\text{II}} \cos \alpha_{tr}}{n_{\text{I}} \cos \alpha_i + n_{\text{II}} \cos \alpha_{tr}} = -\frac{\sin(\alpha_i - \alpha_{tr})}{\sin(\alpha_i + \alpha_{tr})} \quad (3.18a)$$

$$r_{//} = \frac{-n_{\text{II}} \cos \alpha_i + n_{\text{I}} \cos \alpha_{tr}}{n_{\text{I}} \cos \alpha_{tr} + n_{\text{II}} \cos \alpha_i} = -\frac{\tan(\alpha_i - \alpha_{tr})}{\tan(\alpha_i + \alpha_{tr})} \quad (3.18b)$$

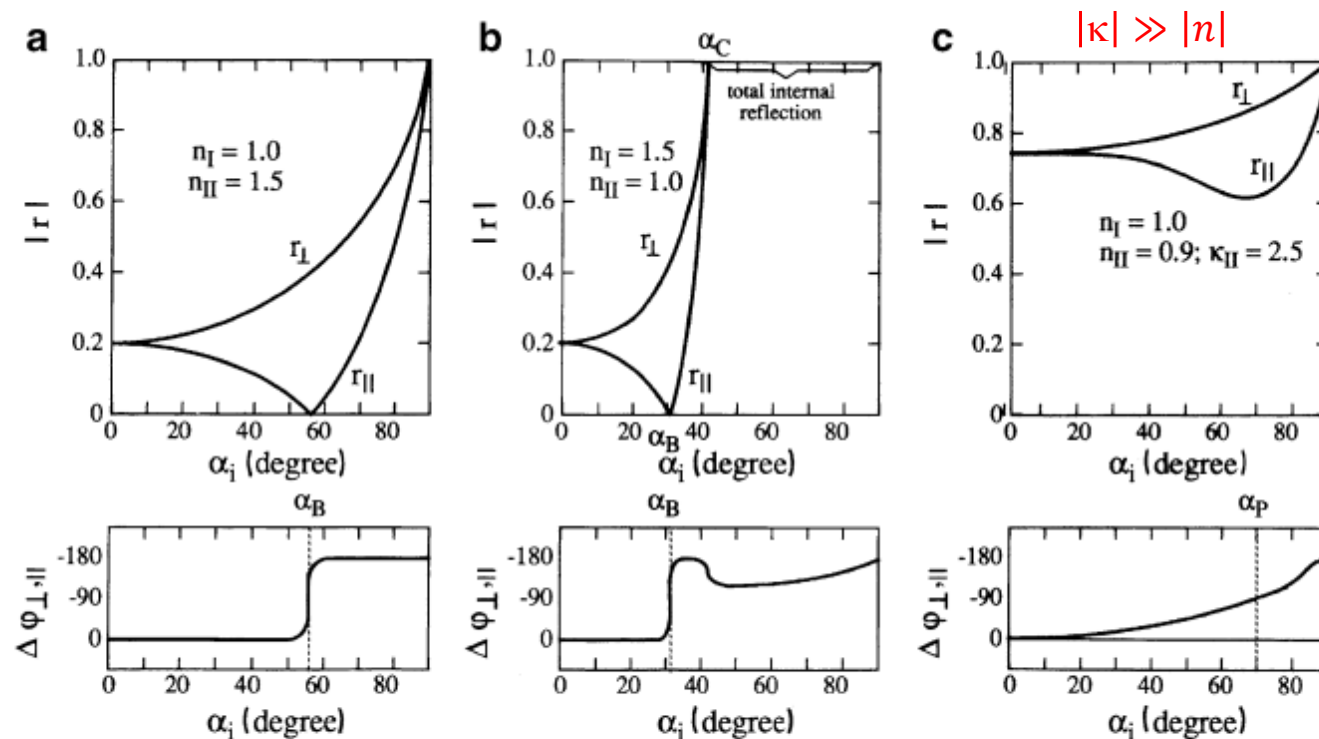
$$t_{\perp} = \frac{2n_{\text{I}} \cos \alpha_i}{n_{\text{I}} \cos \alpha_i + n_{\text{II}} \cos \alpha_{tr}} = \frac{2 \sin \alpha_{tr} \cos \alpha_i}{\sin(\alpha_i + \alpha_{tr})} \quad (3.18c)$$

$$t_{//} = \frac{2n_{\text{I}} \cos \alpha_i}{n_{\text{I}} \cos \alpha_{tr} + n_{\text{II}} \cos \alpha_i} = \frac{2 \sin \alpha_{tr} \cos \alpha_i}{\sin(\alpha_i + \alpha_{tr}) \cos(\alpha_i - \alpha_{tr})} \quad (3.18d)$$

- From (3.18):

$$r_{//,\perp}(\alpha_i = 0) = \frac{n_{\text{I}} - n_{\text{II}}}{n_{\text{I}} + n_{\text{II}}}; t_{//,\perp}(\alpha_i = 0) = \frac{2n_{\text{I}}}{n_{\text{I}} + n_{\text{II}}} \quad (3.18e)$$

- Fig.3.6 formulae graphically and the phase shift between the various reflected components.
- For the experimentally accessible quantities R and T



Strong absorption is not covered by Eq (3.18)

Fig. 3.6 The modulus of the reflection coefficient r for the (electric) field amplitudes according to (3.18) and the phase difference between the reflected amplitudes for the orientations $r_{\perp}$ and $r_{\parallel}$, and reflection at an optically thicker (a) and thinner (b) medium and for reflection at a strongly absorbing medium (c) (According to [76P1])

- For the $R_{\perp}$ increases monotonically with α_i
- For weak absorption ($n \gg k$):

$$R(\alpha_i = 0) = \left(\frac{n_{II} - n_I}{n_{II} + n_I} \right)^2 \quad (3.19a)$$

- For $n_I = 1$, again for weak absorption:

$$R(\alpha_i = 0) = \left(\frac{n_{II} - 1}{n_{II} + 1} \right)^2 \quad (3.19b)$$

- For strong absorption and $n_I = 1$:

$$R(\alpha_i = 0) = \frac{(n_{II} - 1)^2 + k_{II}^2}{(n_{II} + 1)^2 + k_{II}^2} \quad (3.19c)$$

- While for grazing incidence in all cases:

$$R(\alpha_i = 90^\circ) = 1 \quad (3.19d)$$

- In contrast $R_{//}$ goes through a minimum at a certain angle (weak absorption):

$$R_{//}(\alpha = \alpha_B) = 0 \quad (3.20)$$

1. Brewster's angle

2. Only the perpendicularly to the plane of incidence is reflected.

3. Can be used to polarize light.

4. The transmitted beam is not strictly polarized.

- From (3.18b), when $r_{//} = 0$:

$$n_{II} \cos \alpha_i = n_I \cos \alpha_{tr} \quad \text{or} \quad \tan(\alpha_i + \alpha_{tr}) = \infty \quad (3.21)$$

- The nontrivial solution: $\alpha_i + \alpha_{tr} = 90^\circ$ (3.22)

- For $n_{II} < n_I$, the critical angle $\cos \alpha_c$ for total internal reflection exists.

- For strong absorption, R does not reach zero for any polarization .
- The definition of α_p : 1. when the slopes of the curves $R_{//}(\alpha_i)$ and $R_{\perp}(\alpha_i)$ are equal
2.The minimum of $R_{\perp}$
- For a lossless interface, the power (energy per unit of time): $P_i = P_r + P_{tr}$ (3.23a)
- $|t_{//,\perp}|^2, |r_{//,\perp}|^2$ give the energy flux density.

- The transmitted one as show in Fig.3.7 :

$$T_{//,\perp} = \frac{P_{tr}}{P_i} = \frac{I_{tr} \cos \alpha_{tr}}{I_i \cos \alpha_i} = |t_{//,\perp}|^2 \frac{\cos \alpha_{tr}}{\cos \alpha_i}$$

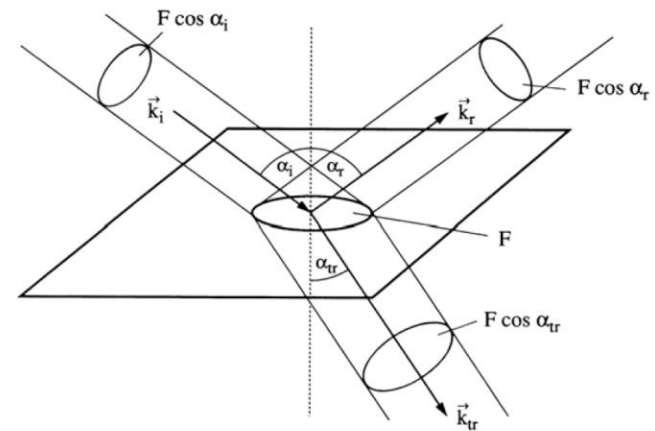


Fig. 3.7. The relation of the cross sections of incident, transmitted and reflected beams at a plane interface between two media

3.1.5 Extinction and Absorption of Light

- Now we discuss on the propagation of a beam within a medium.
- Assumptions: a strictly parallel beam and neglect all diffraction losses .
- The decay usual form: $I = I_0 e^{-\alpha(\omega)d}$ Lambert' law (3.24)
- When the materials active : $\alpha < 0$,while passive : $\alpha > 0$
- The attenuation in 3.24 comprises : $\alpha_{extinction}(\omega) = \alpha_{absorption}(\omega) + \alpha_{scattering}(\omega)$ (3.25)

1.Absorption: transform into like heat , chemical energy or electromagnetic radiation
 2.Scattering: The un-shifted component and the frequency shifted parts .

- If the particles are small compared to λ : $\frac{I_{scatter}}{I_{incident}} \propto \omega^4$ (3.26)
 Eg. The sunlight propagating through clear atmosphere .
- If the scattering or absorbing particles are diluted (Beer's law or Lambert-Beer's law):

$$\alpha(\omega) = n_p \alpha_s(\omega) \quad (3.27)$$
 α_s is the specific extinction constant.

The relationship of the **concentration of a substance** in a media and the **amount of light it absorbs**

3.1.6 Transmission Through a Slab of Matter and Fabry Perot Modes

- The transmission and reflection of a plane-parallel slab of media of geometrical thickness d with ideal, lossless surfaces.

- For strong absorption**, easily detect the beam reflected from the front surface:

$$\hat{R}(\omega) = R(\omega) \quad \text{for} \quad \alpha(\omega)d \gg 1$$

And the most regime : $1 \leq \alpha(\omega)d \leq 5$

- $$\hat{T}(\omega) = [1 - R^{I \rightarrow II}(\omega, \alpha_i)] \exp[-\alpha(\omega)d \cos^{-1} \alpha_{tr}] \cdot [1 - R^{II \rightarrow I}(\omega, \alpha_{tr})] \quad (3.20a)$$

$$\hat{T} \approx [1 - R(\omega)]^2 e^{-\alpha(\omega)d} \quad \text{for normal incidence} \quad (3.20b)$$

- Reflection for normal incidence: $\hat{R}(\omega) \approx R(\omega) + [1 - R(\omega)]^2 R(\omega) e^{-2\alpha(\omega)d} \approx R(\omega) \quad (3.29c)$

- For weak absorption for $\alpha(\omega)d \leq 1$, we must consider multiple reflection. **1. For $d l_c^{-1} \gg 1$**

$$\hat{T}(\omega) \approx \frac{[1 - R(\omega)]^2 e^{-\alpha(\omega)d}}{1 - R(\omega)^2 e^{-2\alpha(\omega)d}} \approx [1 - R(\omega)]^2 e^{-\alpha(\omega)d} \quad \text{for} \quad ad \leq 1 \quad (3.29d)$$

2. For long coherence lengths, form a Fabry-Perot resonator, **Case I:**

$$\lambda_m = \frac{d 2n(\omega)}{m} \quad \text{or} \quad \omega_m = \frac{m \pi c}{d n(\omega_m)} \quad \delta = n(\omega) k^{vac} d = n(\omega) \frac{\omega d}{c} = m \pi, m=1,2,3... \quad (3.30)$$

All partial waves reflected at the two interfaces interfere constructively in the slab,

$$\hat{T} \rightarrow 1, \hat{R} \rightarrow 0$$

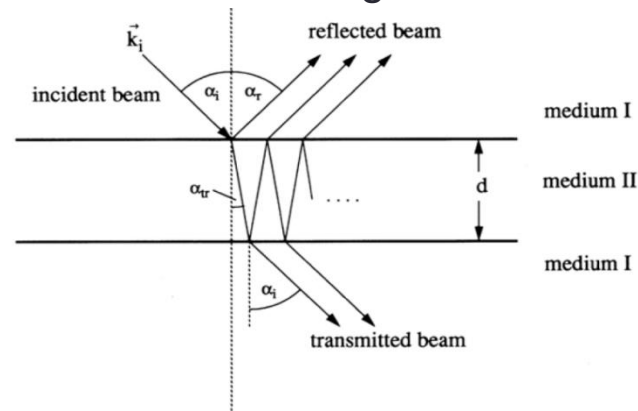


Fig. 3.8. Multiple reflection of an incident light beam in a plane-parallel slab of matter

Case II: All partial waves interfere destructively in the resonator

$$\hat{R} \leq 1, \hat{T} \ll 1$$

- Application: Etalon or Fabry-Perot resonator .
- The general formula for the FP resonator : $\hat{T} = \frac{A}{1 + F \sin^2 \delta}, \hat{R} = \frac{B + F \sin^2 \delta}{1 + F \sin^2 \delta}$ (3.31a)

$$\text{with } F = \frac{4R_\alpha}{(1 - R_\alpha)^2} \quad A = \frac{e^{-\alpha d}(1 - R_F)(1 - R_B)}{(1 - R_\alpha)^2} \quad B = \frac{R_F \left(1 - \frac{R_\alpha}{R_F}\right)}{(1 - R_\alpha)^2}$$

$$R_\alpha = (R_F R_B)^{\frac{1}{2}} e^{-\alpha d} \quad \delta = n(\omega) k^{vac} d = n(\omega) \frac{\omega d}{c}$$

Often $R_F = R_B = R$, and when $\alpha d = 0$, we get $A = 1, B = 0$

- For $\alpha = 0$ and normal incidence

$$\hat{T} = \frac{A}{1 + F' \sin^2 \delta}, \hat{R} = \frac{F' \sin^2 \delta}{1 + F' \sin^2 \delta} = 1 - \hat{T} \quad (3.32a,b)$$

$$F' = \frac{4R}{(1 - R)^2} \quad (3.32c)$$

1. For vanishing damping $\alpha d = 0$, $\hat{T}(\omega)$ reaches unity and $\hat{R}(\omega)$ reaches zero
2. Increase F' makes the FP resonances narrower.
3. $\hat{T} + \hat{R} = 1$
4. For finite damping, the height of the resonance decreases with increasing αd

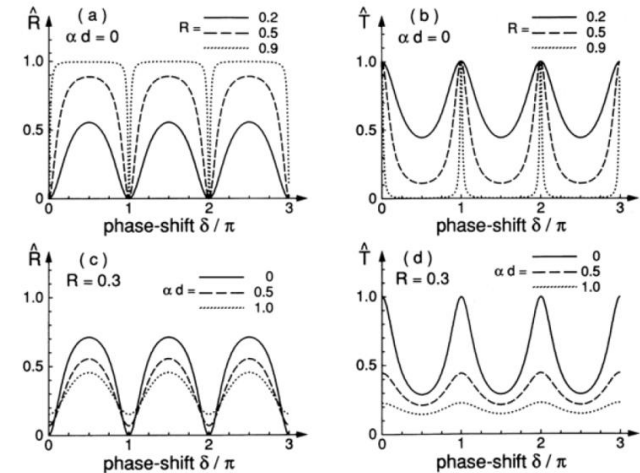


Fig. 3.9. Transmission $\hat{T}$ (b, d) through and reflection $\hat{R}$ (a, c) from a Fabry-Perot resonator as a function of the phase shift δ for one half round trip for various values of the reflectivity R of a single surface or the optical density αd of the medium in the resonator

• 3.1.7 Birefringence and Dichroism

- An anisotropic material: hexagonal wurtzite structure of several semiconductors. In such situations, the dielectric function $\varepsilon(\omega)$ is generally a tensor.

$$D_i = \varepsilon_0 \sum_j \varepsilon_{ij} E_j; \quad i, j = x, y, z \quad (3.33)$$

In principle ε – *tensor* has nine components. Consider $\varepsilon_{ij} = \varepsilon_{ji}$, left six independent. By a suitable rotation of the cartesian coordinate system, left three elements.

$$\varepsilon_{xx}(\omega) = \varepsilon_{yy}(\omega) \neq \varepsilon_{zz}(\omega) \quad (3.34a)$$

For biaxial systems of lower symmetry : $\varepsilon_{xx}(\omega) \neq \varepsilon_{yy}(\omega) \neq \varepsilon_{zz}(\omega) \neq \varepsilon_{xx}(\omega)$ (3.34b)

- For cubic crystals: for $\mathbf{k} = 0$: $\varepsilon_{xx}(\omega) = \varepsilon_{yy}(\omega) = \varepsilon_{zz}(\omega)$ (treated as a scalar quantity)
- for $\mathbf{k} \neq 0$, birefringence and dichroism may appear.
- The consequences tensor character of $\varepsilon(\omega)$ are birefringence and dichroism. See F.3.10

The dichroitic region sometimes covers a wide spectral range, some cases the whole visible part. eg. polaroid films contain long organic molecules.

1. Dichroitic material must show birefringence
2. Birefringent material must have some spectral range in which dichroism occurs.

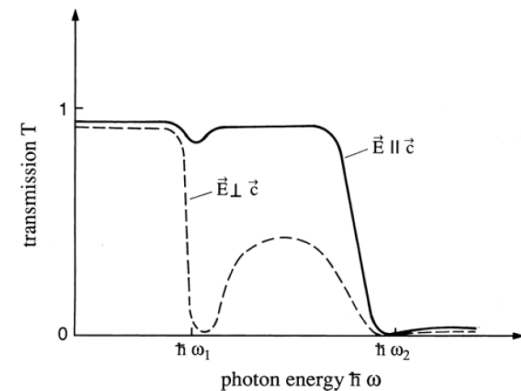


Fig. 3.10. Schematic drawing of the transmission spectra of a dichroitic material for two different polarizations of light with respect to the crystallographic axis

Dichroism means literally that a crystal has two different colors depending on the direction of observation. In a more general sense one describes with the word dichroism every dependence of the absorption spectra on the direction of polarization.

- See Fig.3.11, the ordinary (o) beam and the extraordinary (eo) beam.
- Ordinary (o) beam: perpendicular to the main section.
- extraordinary (eo) beam: has $\mathbf{E} \parallel \mathbf{c}$ and $\mathbf{E} \perp \mathbf{c}$
- Application: use birefringent materials as polarizers.
- Negative-uniaxial birefringence: $\Delta n := n_{eo} - n_o < 0$
- Positive-uniaxial birefringence: $\Delta n := n_{eo} - n_o > 0$
- Two simple situations: 1. $\mathbf{k} \parallel \mathbf{c}$ observes the ordinary beam only
2. $\mathbf{k} \perp \mathbf{c}$ By a polarizer, choose the ordinary or extraordinary.
- Oblique incidence on a surface (see Fig.3.12) are much more complicated.
- Two rather simple ways to present birefringence:

1. The indicatrix
2. The phase velocity in polar coordinates.

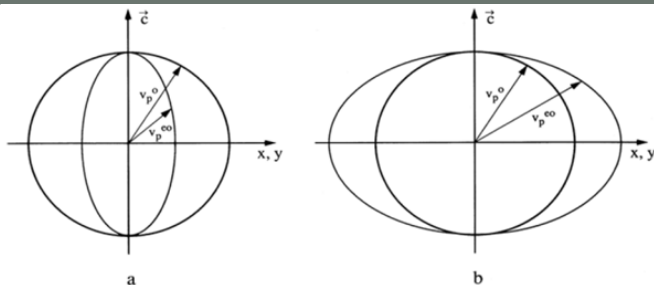


Fig. 3.13. The phase velocity of light in an optically uniaxial material shown for one frequency ω in polar coordinates for positive (a) and negative (b) birefringence for the ordinary (o) and extraordinary (eo) beams

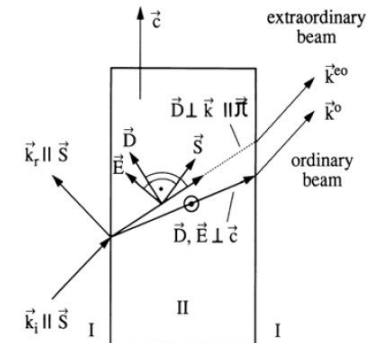


Fig. 3.11. Birefringence for an unpolarized beam falling on a birefringent material at a given angle with the crystallographic axis $\mathbf{c}$ parallel to the interface

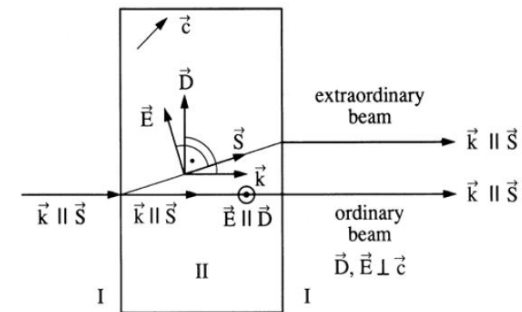


Fig. 3.12. Same as Fig. 3.11 but for normal incidence on a surface cut under an arbitrary angle with respect to $\mathbf{c}$

• 3.1.8 Optical Activity (旋光性)

- An optically active material splits a linearly polarized incident beam into a left σ^{-1} and right σ^{+1} circularly polarized one with equal amplitude, both components travel with different phase velocities. A superposition after a certain distance results again in linearly polarized light, but with a tilted direction of polarization
- Occurs in amorphous solids including liquids and gases. Also occurs in some crystalline solids: quartz or wurtzite-type crystals .

$$\delta_{\text{rot}} = \varrho(\omega)l \quad (3.35a)$$

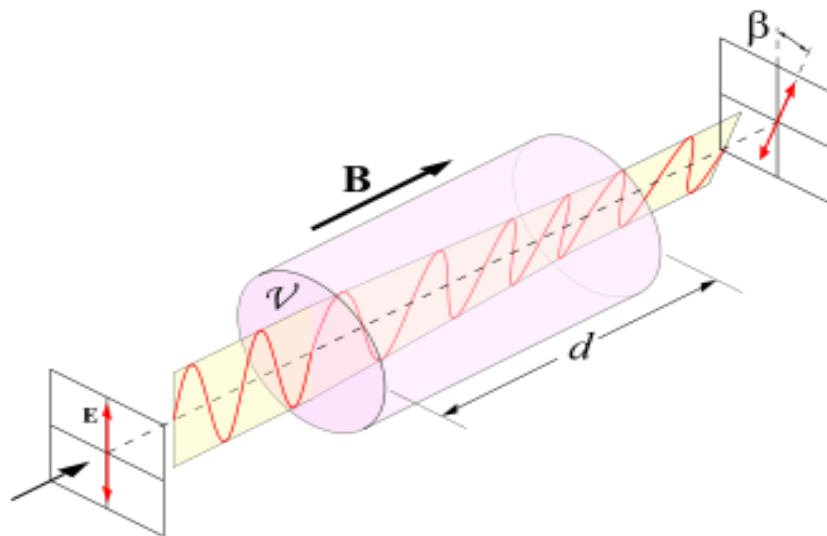
$$\varrho(\omega) = \frac{\pi}{\lambda_0}(n_- - n_+) \quad (3.35b)$$

$$\begin{pmatrix} D_1 \\ D_2 \\ D_3 \end{pmatrix} = \varepsilon_0 \begin{pmatrix} n^2 & -iG & 0 \\ iG & n^2 & 0 \\ 0 & 0 & n^2 \end{pmatrix} \begin{pmatrix} E_1 \\ E_2 \\ E_3 \end{pmatrix} \quad (3.35c)$$

$$n_{\pm} = \sqrt{n^2 \pm G} \quad (3.35d)$$

- The Faraday effect: The optical activity induced by a magnetic field

$$S_+ = V(\omega)lB \quad (3.35e)$$



The Faraday effect

- 3.2 Microscopic Aspects
- 3.2.1 Absorption, stimulated and spontaneous emission, virtual excitation

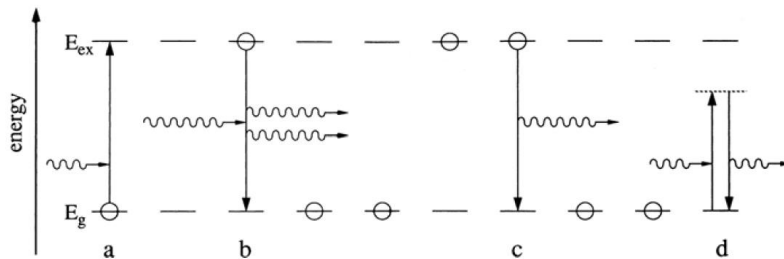


Fig. 3.16. Schematic representations of various interaction processes of light with matter namely absorption (a), stimulated emission (b), spontaneous emission (c) and virtual excitation (d)

- Two-level “atoms”

Fig.3.16a: Absorption process $\hbar\omega = E_{ex} - E_g$

Fig.3.16b: Stimulated emission,

Basic mechanism for all lasers (light **a**mplification by **s**timulated **e**mission of **r**adiation).

Fig.3.16c: Spontaneous emission, an electron in the excited state transit to the ground state by itself.

Fig.3.16d: Virtual excitation: with an energy different from the eigen-energy of this excited state.

Uncertainty principle of quantum mechanics :

$$\Delta x_i \Delta p_i \geq \hbar \quad \text{for } i = 1, 2, 3. \quad (3.37a)$$

$$\Delta E \Delta t \approx \hbar \quad (3.37b)$$

From 3.37b , it is possible to violate energy conservation by an amount ΔE up until a maximum time Δt .

• 3.2.2 Perturbative Treatment of the Linear Interaction of light with Matter

The Hamiltonian of the system elaborating the perturbation terms.

How a perturbation causes transitions between various eigenstates.

Join these two things together ending with an understanding of the theoretical description of absorption and stimulated or spontaneous transitions.

• The Hamiltonian of the total system: $H = H_{el} + H_{rad} + H_{interac}$ (3.40)

• Treat the radiation field as a small perturbation.

• Replacing the conjugate momentum $\mathbf{p}$ by: $\mathbf{p} \rightarrow \mathbf{p} - e\mathbf{A}$

if we replace $\mathbf{p}$ by its operator: $\mathbf{p} = \frac{\hbar}{i}\nabla$

then the single particle Hamiltonian reads: $H = \frac{1}{2m} \left(\frac{\hbar}{i}\nabla - e\mathbf{A} \right)^2 + V(\mathbf{r})$ (3.43)

Using the Coulomb gauge : $H = \frac{\hbar^2}{2m}\nabla^2 + V(\mathbf{r}) - \frac{e}{m}\mathbf{A}\frac{\hbar}{i}\nabla + \frac{e^2}{2m}\mathbf{A}^2$

$$H = H_{el} - \frac{e}{m}\mathbf{A}\frac{\hbar}{i}\nabla + \frac{e^2}{2m}\mathbf{A}^2 \qquad H = H_{el} + H^{(1)} + H^{(2)}$$

- From the above, if we assume the light intensity are small, $H^{(1)}$ is a perturbation term of first order and $H^{(2)}$ is small of second order.
- Fermi's golden rule for the transition rate ω_{ij} :

Time-dependent Schrodinger equation : $H\psi = i\hbar \frac{\partial \psi}{\partial t}$
 with the stationary solutions when $H = H_0$ $\psi_n(\mathbf{r}, t) = \varphi_n(\mathbf{r})e^{-i\left(\frac{E_n}{\hbar}\right)t}$ (3.47b)

For with a perturbation $H^{(1)}$:

$$\psi(\mathbf{r}, t) = \sum_n a_n(t) \varphi_n(\mathbf{r}) e^{-i\left(\frac{E_n}{\hbar}\right)t} \quad (3.48)$$

Assume the perturbation is switched on at $t = 0$. And:

$$\left. \begin{array}{l} a_i(t) = 1 \\ a_{n \neq i}(t) = 0 \end{array} \right\} \text{ for } t \leq 0 \quad (3.49)$$

For $t > 0$, **transition rate - Fermi golden rule**

$$\omega_{ij} = \frac{2\pi}{\hbar} |H_{ij}^{(1)}|^2 D(E) \quad (3.50a)$$

The transition matrix element given by:

$$H_{ij}^{(1)} = \int \psi_j^*(\mathbf{r}) H^{(1)} \psi_i(\mathbf{r}) d\tau = \langle \psi_j | H^{(1)} | \psi_i \rangle \quad \omega_{ij} \propto |H_{ij}^{(1)}|^2 \quad (3.50b)$$

- If $H_{ij}^{(1)} = 0$, then the second-order contribution reads:

$$\omega_{ij}^{(2)} = \frac{2\pi}{\hbar} \left| \sum_{k \neq i, j} \frac{H_{jk}^{(1)} H_{ki}^{(1)}}{E_i - E_k} + H_{ij}^{(2)} \right|^2 D(E) \quad (3.51)$$

- Some more detail for the perturbations:

with the vector potential $\mathbf{A}$: $\mathbf{A} = \mathbf{A}_0 e^{i(\mathbf{k}\mathbf{r} - \omega t)}$ (3.52a)

we find , e.g., for the absorption process (3.50):

$$\begin{aligned} \omega_{g \rightarrow ex} &= \frac{2\pi}{\hbar} \left| \frac{-e\hbar}{im} \mathbf{A}_0 \cdot \int \varphi_{ex}^*(\mathbf{r}) e^{i(E_{ex}/\hbar)t} \mathbf{e}_A e^{i(\mathbf{k}\mathbf{r} - \omega t)} \nabla \varphi_g(\mathbf{r}) e^{-i(E_g/\hbar)t} d\tau \right|^2 D(E_g + \hbar\omega) \\ &\propto \left| A_0 \left\langle \varphi_{ex} \left| H^{(1)} \right| \varphi_g \right\rangle \right|^2 D(E_g + \hbar\omega) =: \\ &A_0^2 \left| H_{eg}^{(1)} \right|^2 D(E_g + \hbar\omega) \end{aligned} \quad (3.52b)$$

A significant transition rate occurs only if the time dependent vanish :

$$E_{ex} - E_g - \hbar\omega = 0 \quad (3.53a)$$

If the φ_i have a wave vector $\mathbf{k}$: $\hbar\mathbf{k}_{ex} - \hbar\mathbf{k}_g - \hbar\mathbf{k} = 0$ (3.53b)

$$\text{so : } \omega_{ij} \propto A_0^2 \left| H_{eg}^{(1)} \right|^2 \propto I \left| H_{eg}^{(1)} \right|^2 \propto N_{ph} \left| H_{eg}^{(1)} \right|^2 \quad (3.54)$$

$$\text{Because of the orthonormal set: } \left| \int \varphi_j^* H^{(1)} \varphi_i d\tau \right|^2 = \left| \int \varphi_i^* H^{(1)} \varphi_j d\tau \right|^2 \quad (3.55)$$

So we find that:

The probabilities for induced emission and absorption are the same and that the rates differ only by factors containing the number of atoms in the upper and lower states.

- Simplify the interaction operator $H^{(1)}$:

The dipole approximation:

Thus expand $e^{i\mathbf{k}\mathbf{r}}$ in (3.52) : $e^{i\mathbf{k}\mathbf{r}} = 1 + \frac{i\mathbf{k}\mathbf{r}}{1!} + \frac{i\mathbf{k}\mathbf{r}^2}{2!} + \dots \simeq 1$.

Also:

$$\hbar\mathbf{k}_{ex} - \hbar\mathbf{k}_g - \hbar\mathbf{k} = 0 \rightarrow \hbar\mathbf{k}_{ex} - \hbar\mathbf{k}_g = 0$$

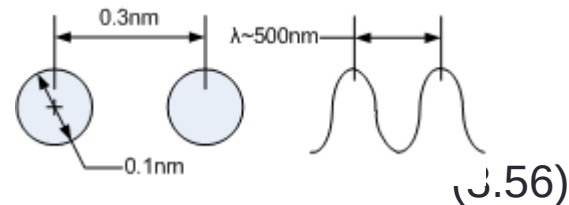
with : $H_{ij}^{(1)} \sim \langle \varphi_i | \mathbf{p} | \varphi_j \rangle = \langle \mathbf{p}_{i,j} \rangle$. $\frac{\hbar}{i} \nabla = p = m\dot{\mathbf{r}}$

so we find that :
$$\int \varphi_j^* \frac{\hbar}{i} \nabla \varphi_i d\tau = m \frac{i}{\hbar} (E_i - E_j) \int \varphi_j^* \mathbf{r} \varphi_i d\tau = m\omega \int \varphi_j^* \mathbf{r} \varphi_i d\tau \quad (3.58)$$

Then the transition rate using this so-called dipole approximation :

$$\omega_{ij} \sim I\omega^2 |e_A \langle \mathbf{r}_{ij} \rangle|^2 D(E) = I\omega^2 |H_{ij}^D|^2 D(E) \quad (3.59)$$

$|H_{ij}^D|^2$: the dipole-transition probability .



- To conclude this section: the net rate in Fig.3.16a-c :

$$\frac{\partial N_{ph}}{\partial t} = \underbrace{-N_A \alpha_g N_{ph} |H_{g \rightarrow e}^D|^2}_{\text{Absorption}} + \underbrace{N_A (1 - \alpha_g) (1 + N_{ph}) |H_{e \rightarrow g}^D|^2}_{\text{Spontaneous and stimulated emission}} \quad (3.61)$$

In which α_g is a fraction in the ground state, $(1 - \alpha_g)$ are in the excited state.

From (3.55) : $|H_{g \rightarrow e}^D|^2 = |H_{e \rightarrow g}^D|^2 = |H^D|^2$ (3.62)

and hence:

$$\frac{1}{|H^D|^2} \frac{\partial N_{ph}}{\partial t} = \underbrace{N_{ph} N_A (1 - 2\alpha_g)}_{\text{Linearly depends on } N_{ph}} + \underbrace{N_A (1 - \alpha_g)}_{\text{Independent of } N_{ph}} \quad (3.63)$$

Absorption and stimulated emission
 Spontaneous emission

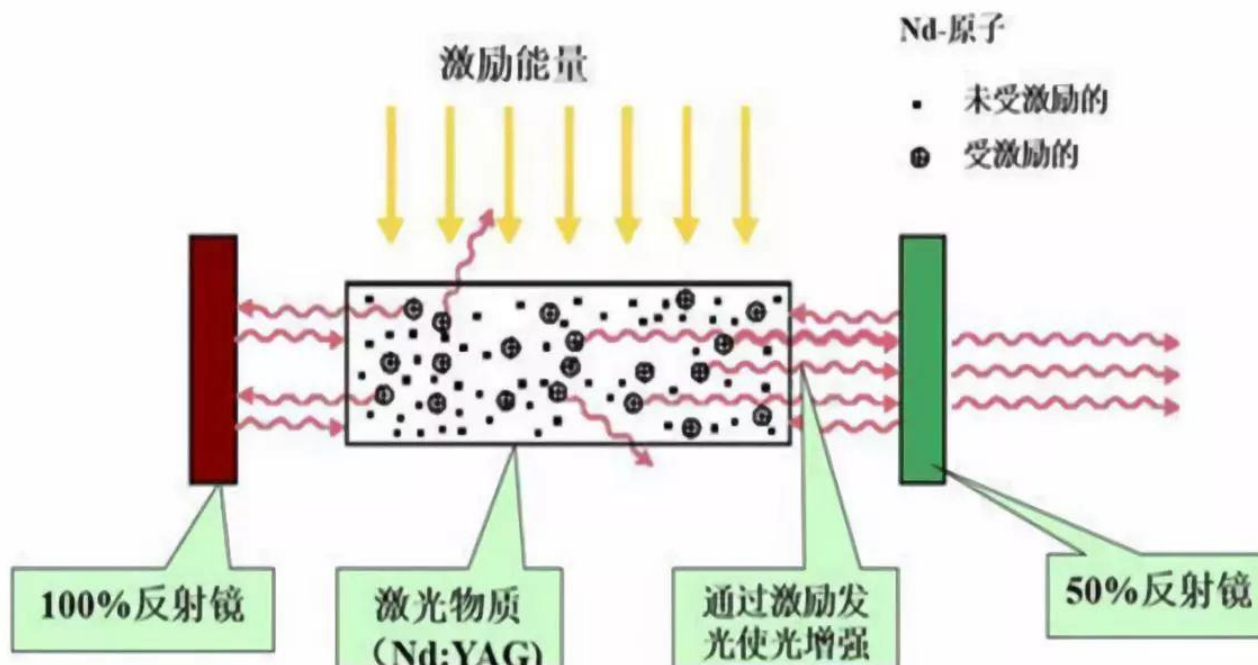
1. For net absorption : $\alpha_g > \frac{1}{2}$
2. Amplification or optical gain : $\alpha_g < \frac{1}{2}$
3. The semiconductor becomes transparent $\alpha_g = \frac{1}{2}$
4. The situation $\alpha_g < \frac{1}{2}$ can't be reached in thermal equilibrium, but only under a pump source

- Elucidated : $\alpha_g(t=0) = 1$, we have absorption : $-\alpha(\omega) \sim N_A(1 - 2\alpha_g)$ (3.64)

so : $t \uparrow, \text{pump power} \uparrow \Rightarrow \alpha_g \downarrow \Rightarrow \alpha(\omega) \downarrow$

when $\alpha_g = \frac{1}{2}$, the absorption vanishes and the material becomes transparent.

Further increase the pump, light amplification achieved.



激光研究简史

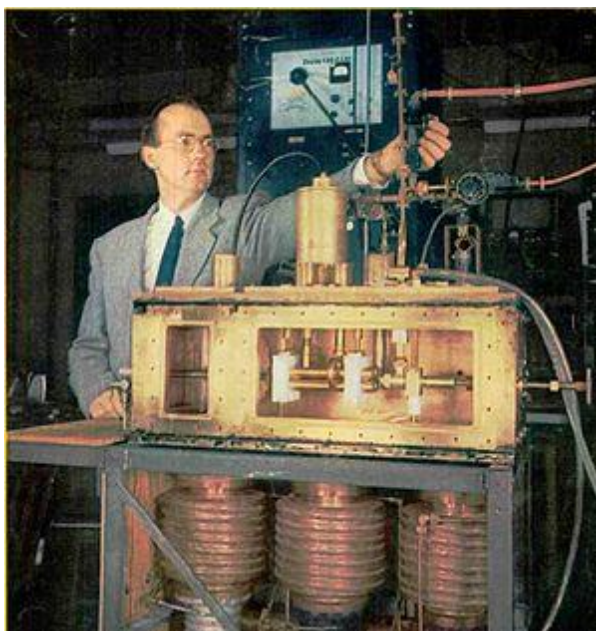
- **1916年**, 爱因斯坦提出了受激辐射的概念,
- **1953年**, 美国物理学家汤斯 (**Charles H. Townes**) 研制成第一台氨分子束微波激光器 (**maser**) 。
- **1955 年**, 苏联的巴索夫和普罗霍洛夫 (**Nikolay Basov and Aleksandr Prokhorov**) 提出光泵浦多能级系统实现粒子数反转的方法 。
- **1957年**, 贝尔实验室的汤斯与肖洛 (**Arthur L. Schawlow**) 提出 **optical maser**, 并于 **1958年** 提交专利申请、在 **Physical Review** 上发表了相关的理论计算。
- **1959年**, 哥伦比亚大学的 **Gordon Gould** 提出了 **laser** 一词 。
- **1960年5月16日**, 美国物理学家西奥多·梅曼 (**Theodore H. Maiman**) 在实验室用一个高强闪光灯管来激发在红宝石水晶里的铬原子, 从而产生一条相当集中的纤细红色光柱 (**694nm**) 。
- **1960年12月**, 美国科学家 **Ali Javan, William R. Bennett, Donal Herriott** (**Bell Labs**) 成功地制造了第一台气体激光器---氦氖激光器 。

In 1964 Charles H. [Townes](#), Nikolay [Basov](#), and Aleksandr [Prokhorov](#) shared the [Nobel Prize in Physics](#), “ for fundamental work in the field of quantum electronics, which has led to the construction of oscillators and amplifiers based on the maser–laser principle ”.



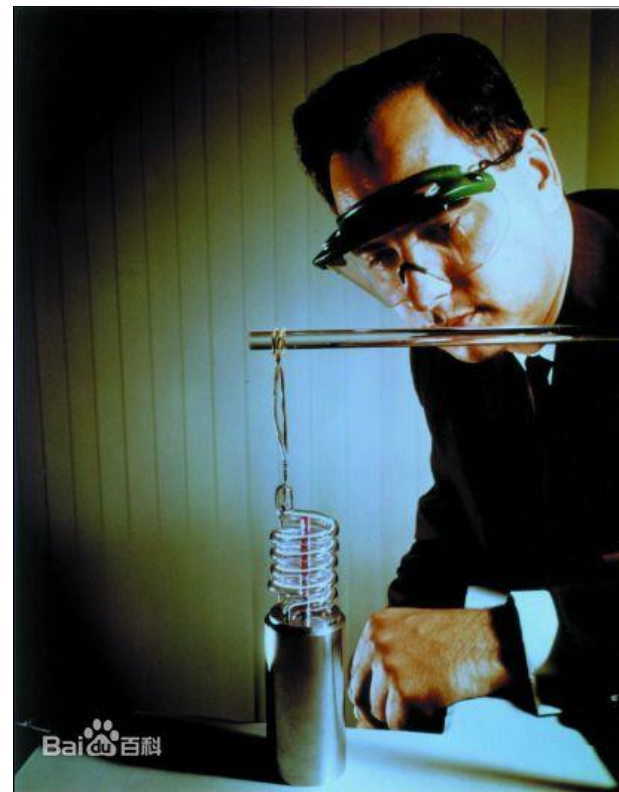
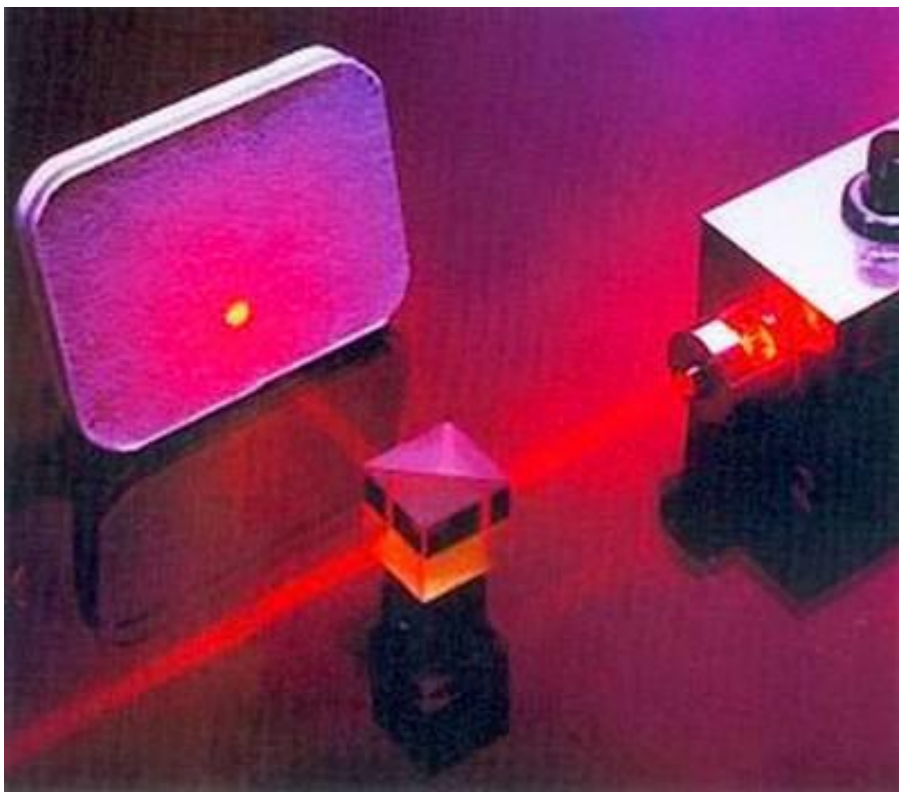
Maser（微波激光器）

- 美国哥伦比亚大学的物理学家查尔斯·汤斯二战时搞过雷达，期望实现微波放大。1954年，汤斯终于把Maser做出来了，虽然放大的是微波，但是这为Laser的发明作好了准备
- 1958年，汤斯又和同事阿瑟·肖洛发现，将氦光灯泡发射的光照在一种稀土晶体上，晶体会发出鲜艳的、始终会聚在一起的强光



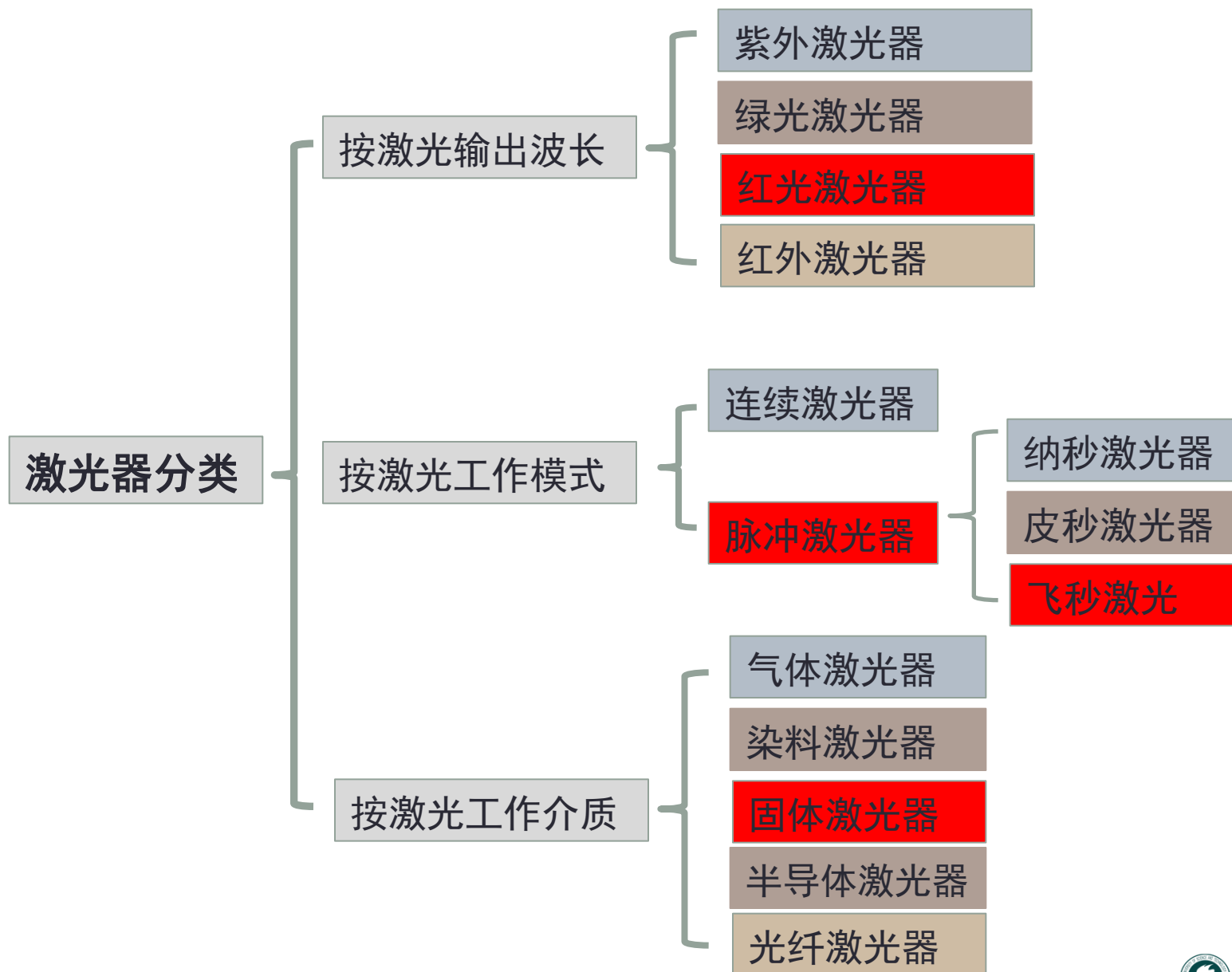
查尔斯·汤斯1964年因激光研究获得诺贝尔奖，阿瑟·肖洛1981年因激光光谱学研究获得诺贝尔奖

第一台激光器的发明



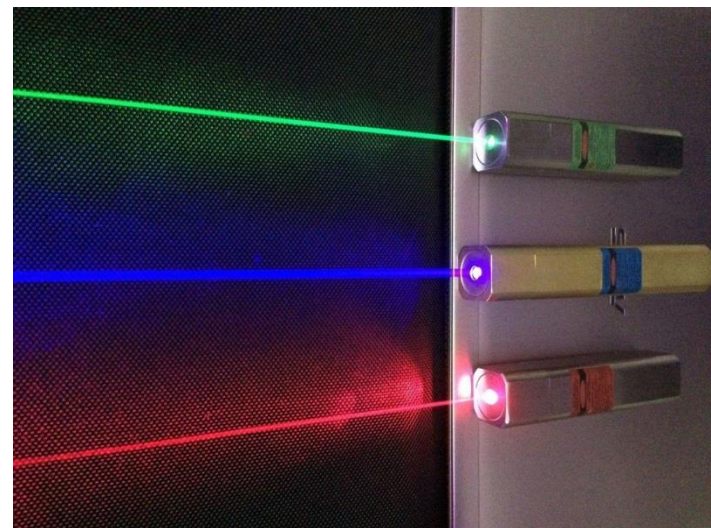
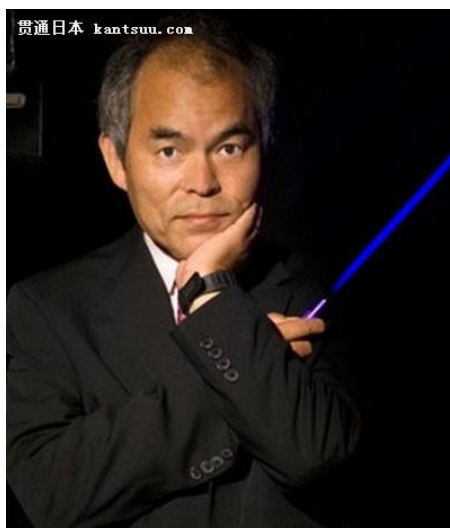
- ❑ 1960年，美国休斯实验室的梅曼用高强闪光灯管来激发**红宝石**。制成了**世界上第一台激光器**
- ❑ 1960年12月，美国科学家Ali Javan, William R. Bennett, Donal Herriott (Bell Labs) 成功地制造了第一台气体激光器——**氦氖激光器**
- ❑ 美国做出激光器后一年，中国马上就由王大珩院士领头在**1961**年做出了**中国第一台激光器**

激光器分类

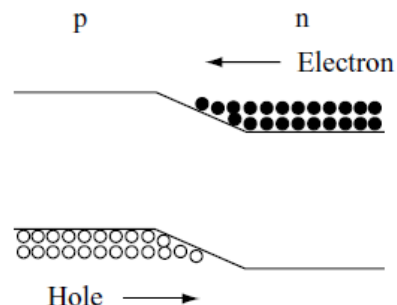
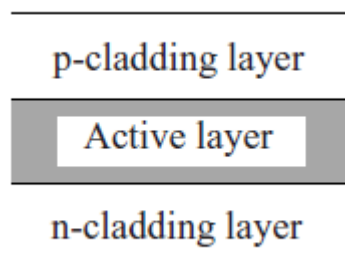


半导体激光器

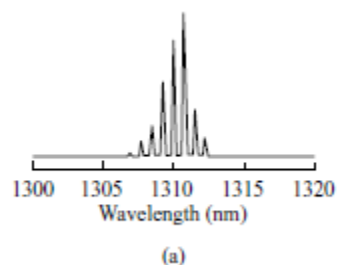
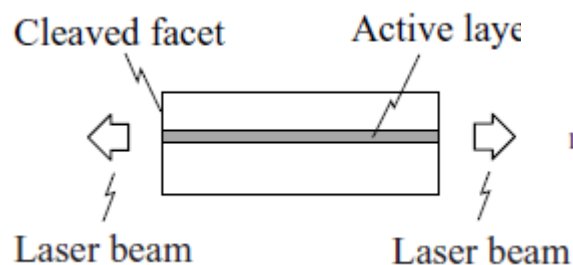
- ❑ 苏联的科学家巴索夫1958年公开发表论文提出了用半导体制造激光器的想法，1961年又发表了“载流子注入”PN结的文章
- ❑ 1962年，GE的Robert N. Hall 和IBM的Marshall Nathan领导的两个研究小组 研制出了GaAs 半导体激光二极管 (850nm)
- ❑ 1962年，GE的 N. Holonyak, Jr. 实现了第一支 GaAs 可见光激光器
- ❑ 1963年，Kroemer 及约飞所的Zh.I. Alferov and R.F. Kazarinov提出异质结结构，为高性能半导体激光器的制作奠定了基础
- ❑ 1970年，Zh. I. Alferov 及贝尔实验室M.B. Panish and I. Hayashi几乎同时报道了第一支室温连续工作的双异质结半导体激光器
- ❑ 1990年代，中村修二发明蓝光和绿光半导体激光器，2014获得诺贝尔奖
- ❑ 半导体激光器电光转换效率高，达50%以上。使用寿命长达10万小时以上，体积小、重量轻、性价比高。应用领域包括光纤通讯、信息存储、激光显示、激光照明、激光加工、**光互联**



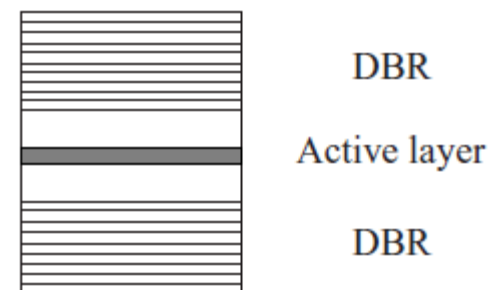
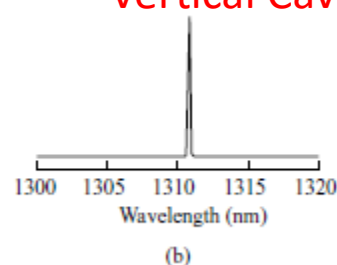
各种类型半导体激光器



Fabry-Perot LD

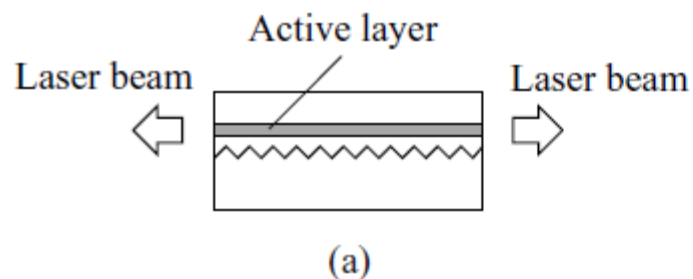


Vertical Cavity Surface Emitting Laser

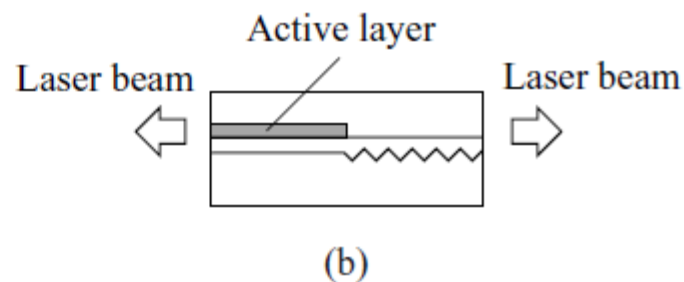


Laser beam

DFB LD



DBR LD



激光核聚变

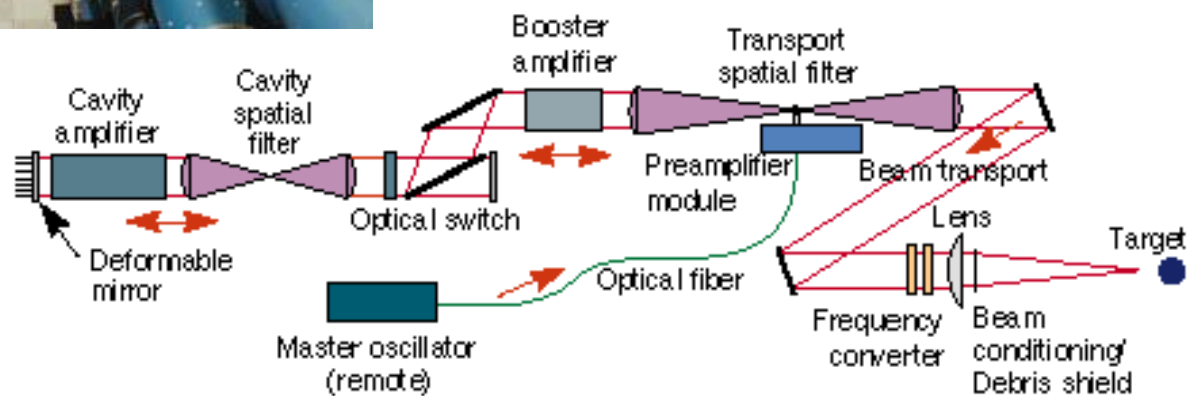
美国国家点火装置



Nova

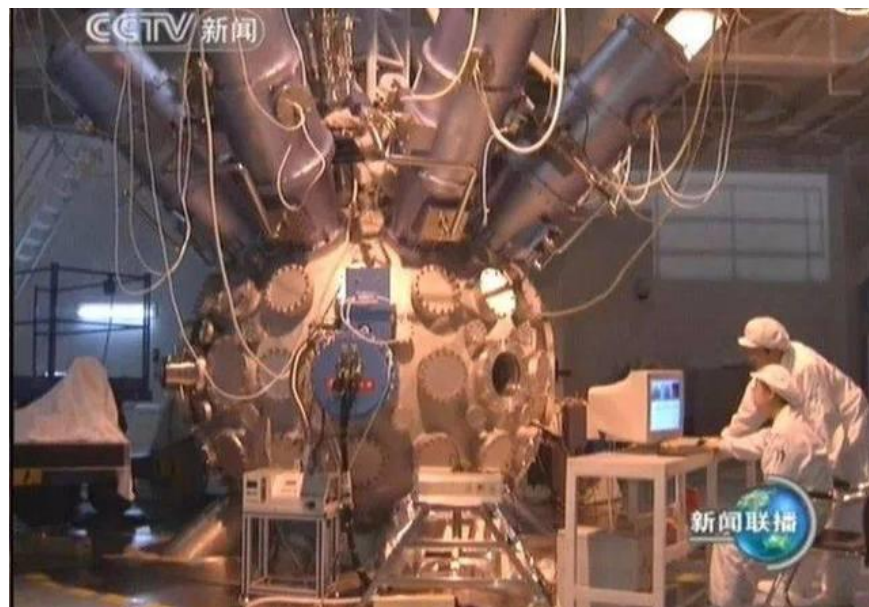


192 shaped pulses; 1.8 MJ total energy



神光装置

- ❑ 中国已经先后研发了神光I、神光II和神光III三代激光实验装置，其中前两个位于上海，而第三个实验装置则建在了四川，与2012年的时候建设完成。根据2004年4月，神光II装置成功突破100万亿瓦（10拍瓦），这意味着神光II在1000万亿分之36秒的超短瞬间内，迸发出相当于全球电网发电总和数十倍的强大功率。这样的杀伤力绝对秒杀任何打击范围内的目标！
- ❑ 神光3是亚洲第一和世界第二的激光装置
- ❑ 神光IV据说已经开始建设
- ❑ 神光装置最终的研究目标是实现激光核聚变



激光在军事领域的应用



中国激光武器



中国“沉默猎手”战术激光器



激光武器打靶破坏效应物

高功率光纤激光器已经广泛应用于军事国防领域！

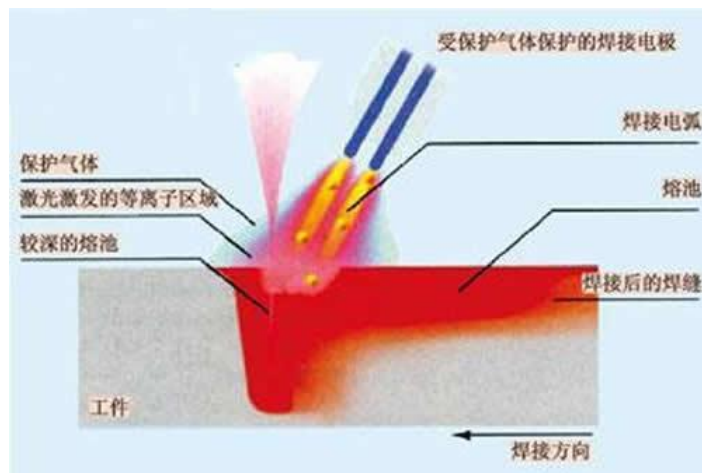


激光焊接

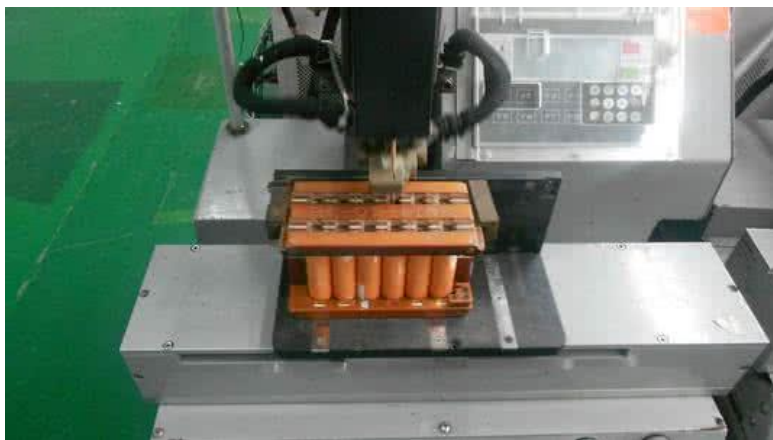
激光焊接具有速度快、深度大、变形小等优点，可焊接难熔材料如钛、石英等，并能对异性材料施焊，效果良好。广泛应用于动力电池、汽车、航空、模具等机械加工领域



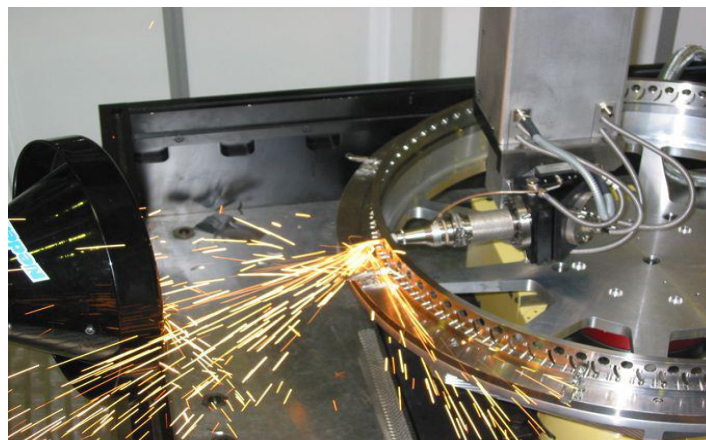
激光焊接机



激光焊原理



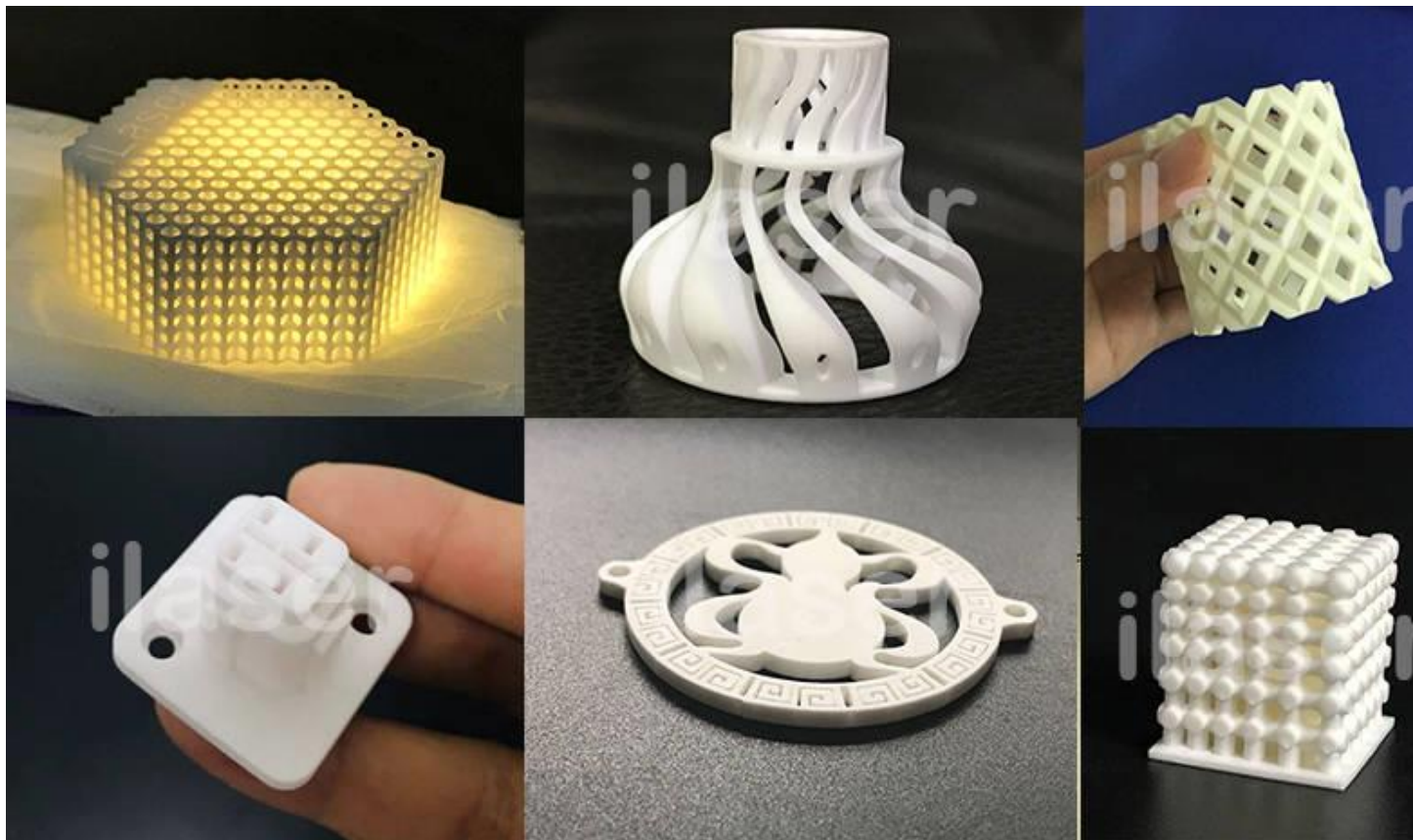
动力电池激光焊接



激光切割焊接一体化加工

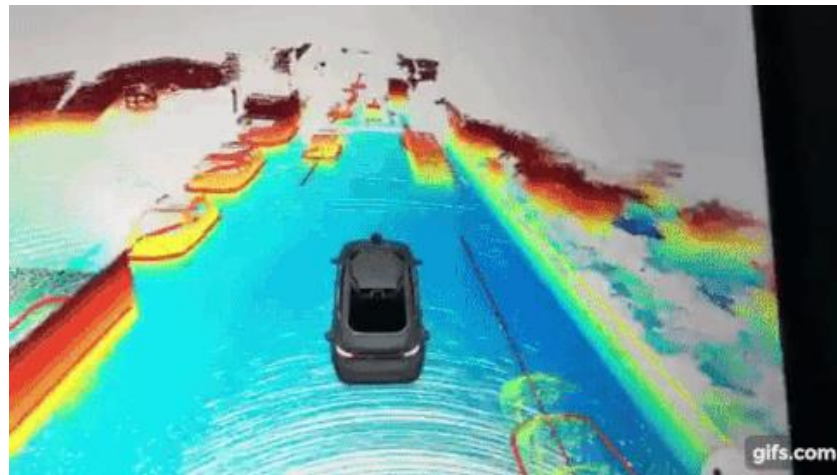
激光3D打印

- ❑ 3D打印技术起源于上个世纪80年代美国，发展迅猛，被认为将引领新一轮工业革命
- ❑ 金属激光3D打印使用高能量的激光束扫描金属粉末床，使金属粉末熔化然后粘结在一起冷却成型进而逐层打印。其优点是允许复杂的几何形状和组件的构建，以前需要多个零件组装而成的部件，通过金属3D打印技术可以简化成单一的而不影响强度或其它机械性能的部件



激光雷达与自动驾驶

38



激光技术未来发展方向

- 更小-更小的体积
- 更高-更高的脉冲能量
- 更快-更短的激光脉冲宽度
- 更强-更强的功率输出

从玻璃芯的褶皱里起身时
它还藏着千万条细碎的光
像被压缩的星河，在密闭的管道里
练习直线的舞蹈 ——
电子与光子撞出火花的瞬间
所有散漫的明亮，突然被拧紧成
一根不容分说的锋芒

它不似晨光会绕开叶片的阻拦
也不像月光总在水面碎成涟漪
一旦挣脱束缚，便带着决绝的笔直
穿过空气时不留声息
落在金属上，就能熔出细小的火焰
在电路板间游走，刻下比发丝更密的纹路
连遥远的星辰，都能被它丈量出距离

偶尔也会收敛锋芒
在舞台的烟雾里，织出流动的彩虹
或是化作医生手中的利刃
在显微镜下，精准地缝合生命的缝隙
它永远保持着纯粹的单色
不掺一丝杂光，却能在不同的领域里
把“专注”写成最耀眼的诗句

